

Activity 0.1.2 What's the Problem?

PLTW

WHAT'S THE PROBLEM?

- ≡ How Can I Help?
- ≡ Engineering Design Process
- ≡ Define the Problem
- ≡ Welcome to Shoe For You!
- ≡ Interview
- ≡ Conclusion

How Can I Help?

GOALS	RESOURCES
• Define a problem based on given data. • Collect information from an interview and apply it to write a detailed problem statement. • Create a detailed design statement based on a problem statement. • Apply ideation techniques to solve a problem.	
GOALS	RESOURCES
• <u>Design Process</u> • <u>Writing a Design Brief</u> • <u>Active Listening and Interpreting</u> • <u>Brainstorming Solutions</u>	

- **Activity 0.1.2 What's the Problem? (Downloadable PDF)**

Engineers spend most of their day designing solutions to help others. They solve problems for people, society in general, and even animals. To begin the activity, see how PLTW alumni Albert Manero noticed a problem in his community and took action.



CAREER CONNECTION

Albert Manero | *Mechanical Engineer*

Inspired by 3D printing as a PLTW student, Albert created his own company to help create prosthetic limbs for children in need.



Reflection



What problems do you notice in your community that an engineer could solve?

Engineering Design Process

Throughout this course, you will encounter many different design problems, just like the newspaper shoes challenge. In that challenge, you likely followed some kind of design process even if you were not aware of it.

However, as you progress through this course, it will be important to follow the Engineering Design Process (Figure 1) to effectively and completely solve problems.

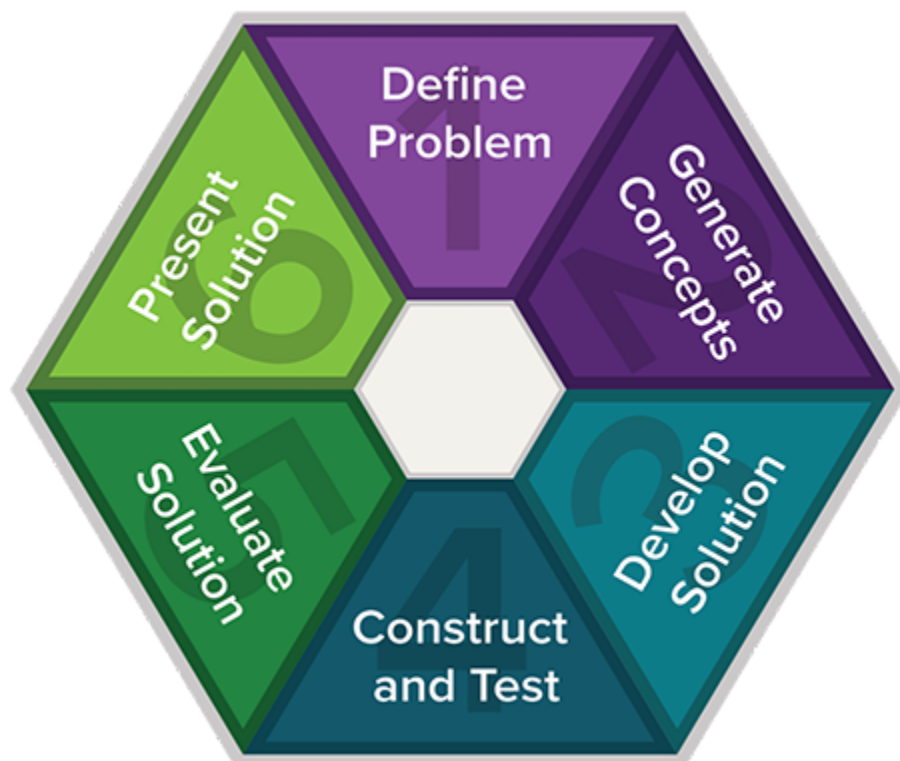


Figure 1. Engineering Design Process

1

Review the [Design Process](#) resource and draw an image of the process for your reference in your notebook.

Define the Problem

As you see in the Engineering Design Process, the first step is to define the problem. Sometimes a problem is simple and requires a simple solution, but other times it can be very complex. To effectively solve a problem for someone, it is crucial to have a clear definition of what the problem is that needs to be solved.

2

Review Define the Problem (located in the Design Process resource) and summarize how to best define a problem.

3

Review the [Writing a Design Brief](#) resource to understand the differences between a problem statement and a design statement.

When solving a problem, it is important to solve the correct problem, and that takes specific effort to determine. One way to get to the root of a problem is by continuing to ask “Why?”

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Examine the following example conversation with two students (Figure 2). Notice how the problem statement

changes as you learn more about the problem when you ask specific “why” questions.

The Five Whys: How to Find the Root Problem

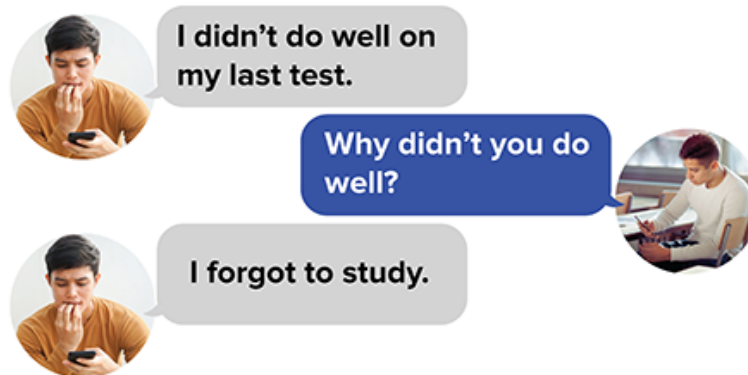


Figure 2. Student Conversation

Initial Problem Statement

Your friend forgets to study for tests.

- ***Why did you forget about the test?***
 - Well, I had another assignment due the same day.
- ***Why didn't you schedule time for both?***
 - I had soccer practice until 5:00 that night.
- ***Why didn't you study after practice?***

- The team went to dinner together after practice.
- ***Why didn't you study after that?***
 - I stayed up too late playing video games the previous night and was too tired and wanted to go to bed.

Final Problem Statement —

Your friend needs to effectively coordinate multiple activities, study time, additional assignments, and other responsibilities so he does not stay up too late and under-perform on a test.

Welcome to Shoe For You!



In the next several activities, you will work directly with a footwear design company, Shoe For You. This company was created to design shoes specifically for you and your needs! You need to choose a client to work with and design a shoe based on their specific needs. Time to meet the clients, Sebastian, Elena, and Kate!

SEBASTIAN	ELENA	KATE
Sebastian is a stay-at-home father. He spends most of his time driving his kids to school, keeping up with their after-school activities, and trying to keep an eye on his youngest daughter who just learned to walk. She likes playing in puddles,		

running, and jumping. He notices how quickly she goes through shoes and wants to create a more durable shoe for her.



SEBASTIAN

ELENA

KATE

Elena is a medical student working on her MD. When she's not in class or studying, she coaches field hockey, plays basketball, and goes rock climbing. Athletic shoes are expensive, and Elena doesn't want to buy a separate pair of shoes for each activity. She is looking for a versatile shoe that she can wear for all three of these sports.



SEBASTIAN

ELENA

KATE

Kate loves running, she runs whenever and wherever she can. She runs at the beach, in her neighborhood, and even on hiking trails.



5

Decide which client you want to work with.

6

Write a problem statement for your chosen client.

Reflection



- Do you think you have enough information to design a shoe for this client?
Why or why not?
- Write a list of questions you still need answered about your client to help you design a better shoe for them.

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Before you move to the next section, review [Active Listening and Interpreting](#) for some useful tips.

Interview

To design the best possible shoe for each client, Shoe For You offers a complementary interview to learn more about their specific needs. This practice is based on [empathy](#) and an understanding that our clients play an integral role in our design process.

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Review the Shoe For You Inbox below to learn more about each client and their specific needs.

Shoe For You Inbox



Sebastian

Reliable shoe for toddler



Elena

Shoe for multiple sports



Kate

Trail running shoes

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Open the [Shoe For You Inbox](#) in a new browser tab.

9

Revise your initial problem statement to include details that were provided in the interviews.

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Write a [design statement](#) with specific [criteria](#) and [constraints](#) for your client based on your revised problem statement.

Now that you have heard all the client interviews, you may wish to work with a different client if another problem interests you more than your initial choice. If you change clients, be sure to write a new problem statement before writing your design statement.

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Choose a brainstorming technique from the [Brainstorming Solutions](#) resource to collect ideas and create three shoe designs to best solve your client's needs based on your design statement.

Conclusion

Question 1

How did your problem statement evolve from your first draft to the second? Why is this important to notice?

Question 2

How did the interview benefit you as a designer?

Question 3

Which brainstorming technique did you choose? Was it effective? How can you improve your brainstorming skills for the next design challenge?